Bayesian Optimization Report: Function 2

GP Health Summary per Iteration

Iteration	GP Health	Status
1	0.742	Good
2	0.741	Good
3	0.743	Good
4	0.746	Good
5	0.740	Good
6	0.741	Good
7	0.743	Good
8	0.742	Good
9	0.741	Good
10	0.740	Good
11	0.740	Good
12	0.740	Good
13	0.741	Good
14	0.740	Good
15	0.739	Good
16	0.737	Good
17	0.737	Good
18	0.737	Good
19	0.736	Good
20	0.736	Good
21	0.735	Good
22	0.726	Good
23	0.735	Good
24	0.718	Good
25	0.717	Good
26	0.724	Good
27	0.723	Good
28	0.715	Good
29	0.723	Good
30	0.723	Good

GP Sigma Summary

Iteration	Mean Sigma	Min Sigma	Max Sigma
1	0.2150	0.1952	0.2331
2	0.1998	0.1686	0.2245
3	0.1980	0.1766	0.2111
4	0.1891	0.1807	0.2006
5	0.1740	0.1379	0.1991
6	0.1703	0.1231	0.1945
7	0.1772	0.1682	0.1891
8	0.1744	0.1656	0.1855
9	0.1725	0.1633	0.1824
10	0.1687	0.1583	0.1790
11	0.1649	0.1544	0.1749
12	0.1612	0.1498	0.1711
13	0.1556	0.1436	0.1665
14	0.1524	0.1388	0.1639
15	0.1506	0.1361	0.1633
16	0.1506	0.1392	0.1590
17	0.1469	0.1344	0.1563
18	0.1455	0.1278	0.1575
19	0.1429	0.1263	0.1547
20	0.1392	0.1220	0.1513
21	0.1374	0.1162	0.1529
22	0.0928	0.0809	0.1235
23	0.1295	0.1078	0.1467
24	0.0980	0.0831	0.1264
25	0.0980	0.0818	0.1267
26	0.0901	0.0763	0.1210
27	0.0876	0.0744	0.1192
28	0.0935	0.0763	0.1224
29	0.0869	0.0723	0.1179
30	0.0860	0.0712	0.1178

Best Results Per Iteration

Iteration	Best Input	Best Output	Kernel	Acquisition	GP Health	Anomalies
1	[0.7798454 0.10698064]	0.3910949235945001	Matern(ν=2.5)	UCB	0.742	None
2	[0.25238643 0.98299864]	0.09305246119775482	Matern(ν=1.5)	UCB	0.741	None
3	[0.62791777 0.96367134]	0.3279870245908535	Matern(ν=1.5)	UCB	0.743	None
4	[0.7454827 0.63471589]	0.4468612995254355	Matern(ν=2.5)	UCB	0.746	None
5	[0.50826488 0.90549457]	0.18343288697720317	Matern(ν=2.5)	UCB	0.740	None
6	[0.81291248 0.71290012]	0.3397789646612033	Matern(ν=2.5)	UCB	0.741	None
7	[0.6843634 0.40957984]	0.4792334049089857	Matern(ν=2.5)	UCB	0.743	None
8	[0.7833206 0.44965124]	0.3815864756179118	Matern(ν=2.5)	UCB	0.742	None
9	[0.62298065 0.57451205]	0.2795257911968283	Matern(ν=2.5)	UCB	0.741	None
10	[0.8362949 0.23679703]	0.2976677527225527	Matern(ν=1.5)	UCB	0.740	None
11	[0.87853145 0.00462042]	0.23670413744094138	Matern(ν=2.5)	UCB	0.740	None
12	[0.00067449 0.04633505]	0.14764068430402563	Matern(ν=2.5)	UCB	0.740	None
13	[0.62718276 0.30579692]	0.29185156518475586	Matern(ν=1.5)	UCB	0.741	None
14	[0.67371647 0.75885233]	0.4519273248208449	Matern(ν=1.5)	UCB	0.740	None
15	[0.78044648 0.85327338]	0.41402523408333614	Matern(ν=1.5)	UCB	0.739	None
16	[0.87051632 0.54082152]	0.25814751998944846	Matern(ν=2.5)	UCB	0.737	None
17	[0.799188 0.97434717]	0.3483640969927279	Matern(ν=2.5)	UCB	0.737	None
18	[0.44182256 0.45461673]	0.227767185877766	Matern(ν=2.5)	UCB	0.737	None
19	[0.42182051 0.98496457]	0.2184495859952162	Matern(ν=2.5)	UCB	0.736	None
20	[0.87286852 0.3309294]	0.25822748557550473	Matern(ν=2.5)	UCB	0.736	None
21	[0.93283563 0.72165494]	0.24786637287854885	Matern(ν=2.5)	UCB	0.735	None

22	[0.48286787 0.08967757]	0.19993261485192915	Matern(ν=2.5)	UCB	0.726	None
23	[0.70902559 0.5372392]	0.5059618618527038	Matern(ν=2.5)	UCB	0.735	None
24	[0.42864213 0.03430564]	0.20827182364036623	Matern(ν=0.5)	UCB	0.718	None
25	[0.90720916 0.01023788]	0.19468614174592228	Matern(ν=0.5)	UCB	0.717	None
26	[0.40475839 0.95174732]	0.1698039077953744	Matern(ν=2.5)	UCB	0.724	None
27	[0.93064649 0.72547917]	0.1732735099180724	Matern(ν=2.5)	UCB	0.723	None
28	[0.49470676 0.72742565]	0.1783185465886769	Matern(ν=0.5)	UCB	0.715	None
29	[0.40949483 0.01709325]	0.1668800304193137	Matern(ν=2.5)	UCB	0.723	None
30	[0.92751329 0.0089832]	0.1689873347190738	Matern(ν=2.5)	UCB	0.723	None

Acquisition Values Per Iteration

Iteration 1

Method	Max Value
UCB	0.788585
EI	0.003644
PI	0.040210
THOMPSON	0.599372
KG	0.003644
ENTROPY	1.214235
PORTFOLIO	0.846484

Iteration 2

Method	Max Value
UCB	0.445852
EI	0.001075
PI	0.015866
THOMPSON	0.566442
KG	0.001075
ENTROPY	1.087996

PORTFOLIO	0.766941

Iteration 3

Method	Max Value
UCB	0.464552
EI	0.001569
PI	0.022800
THOMPSON	0.488972
KG	0.001569
ENTROPY	1.110563
PORTFOLIO	0.746436

Iteration 4

Method	Max Value
UCB	0.497311
EI	0.002936
PI	0.038648
THOMPSON	0.527250
KG	0.002936
ENTROPY	1.113029
PORTFOLIO	0.663776

Iteration 5

Method	Max Value
UCB	0.405572
EI	0.000344
PI	0.006260
THOMPSON	0.495925
KG	0.000344
ENTROPY	1.054837
PORTFOLIO	0.595985

Iteration 6

Method	Max Value
UCB	0.411921

EI	0.000411
PI	0.007309
THOMPSON	0.519453
KG	0.000411
ENTROPY	1.026784
PORTFOLIO	0.561732

Iteration 7

Method	Max Value
UCB	0.506939
EI	0.003123
PI	0.043306
THOMPSON	0.551987
KG	0.003123
ENTROPY	1.114627
PORTFOLIO	0.569271

Iteration 8

Method	Max Value
UCB	0.431117
EI	0.000626
PI	0.010709
THOMPSON	0.477108
KG	0.000626
ENTROPY	1.077513
PORTFOLIO	0.508002

Iteration 9

Method	Max Value
UCB	0.353176
EI	0.000082
PI	0.001694
THOMPSON	0.489891
KG	0.000082
ENTROPY	1.081958

PORTFOLIO	0.456097

Iteration 10

Method	Max Value
UCB	0.370752
EI	0.000111
PI	0.002309
THOMPSON	0.514087
KG	0.000111
ENTROPY	1.015823
PORTFOLIO	0.462171

Iteration 11

Method	Max Value
UCB	0.315144
EI	0.000019
PI	0.000463
THOMPSON	0.430510
KG	0.000019
ENTROPY	1.007801
PORTFOLIO	0.404176

Iteration 12

Method	Max Value
UCB	0.318622
EI	0.000015
PI	0.000378
THOMPSON	0.445165
KG	0.000015
ENTROPY	1.033327
PORTFOLIO	0.419250

Iteration 13

Method	Max Value
UCB	0.344407

EI	0.000024
PI	0.000616
THOMPSON	0.508559
KG	0.000024
ENTROPY	0.940385
PORTFOLIO	0.372882

Iteration 14

Method	Max Value
UCB	0.458252
EI	0.000664
PI	0.013210
THOMPSON	0.473604
KG	0.000664
ENTROPY	0.986089
PORTFOLIO	0.377111

Iteration 15

Method	Max Value
UCB	0.427057
EI	0.000260
PI	0.005779
THOMPSON	0.448165
KG	0.000260
ENTROPY	0.939822
PORTFOLIO	0.425174

Iteration 16

Method	Max Value
UCB	0.314745
EI	0.000007
PI	0.000197
THOMPSON	0.385357
KG	0.000007
ENTROPY	0.951400

PORTFOLIO	0.351174

Iteration 17

Method	Max Value
UCB	0.376052
EI	0.000047
PI	0.001211
THOMPSON	0.368995
KG	0.000047
ENTROPY	0.983844
PORTFOLIO	0.321718

Iteration 18

Method	Max Value
UCB	0.316605
EI	0.000003
PI	0.000100
THOMPSON	0.412858
KG	0.000003
ENTROPY	0.913465
PORTFOLIO	0.299983

Iteration 19

Method	Max Value
UCB	0.306269
EI	0.000002
PI	0.000064
THOMPSON	0.343845
KG	0.000002
ENTROPY	0.899633
PORTFOLIO	0.298223

Iteration 20

Method	Max Value
UCB	0.301871

EI	0.000001
PI	0.000037
THOMPSON	0.371491
KG	0.000001
ENTROPY	0.899914
PORTFOLIO	0.287198

Iteration 21

Method	Max Value
UCB	0.293423
EI	0.000001
PI	0.000029
THOMPSON	0.391123
KG	0.000001
ENTROPY	0.897436
PORTFOLIO	0.276893

Iteration 22

Method	Max Value
UCB	0.216701
EI	0.000000
PI	0.000000
THOMPSON	0.253821
KG	0.000000
ENTROPY	0.768566
PORTFOLIO	0.179471

Iteration 23

Method	Max Value
UCB	0.467421
EI	0.000385
PI	0.010173
THOMPSON	0.534829
KG	0.000385
ENTROPY	0.921859

PORTFOLIO	0.339233

Iteration 24

Method	Max Value
UCB	0.223508
EI	0.000000
PI	0.000000
THOMPSON	0.300389
KG	0.000000
ENTROPY	0.807032
PORTFOLIO	0.186837

Iteration 25

Method	Max Value
UCB	0.210774
EI	0.000000
PI	0.000000
THOMPSON	0.240284
KG	0.000000
ENTROPY	0.785434
PORTFOLIO	0.171227

Iteration 26

Method	Max Value
UCB	0.192410
EI	0.000000
PI	0.000000
THOMPSON	0.233461
KG	0.000000
ENTROPY	0.746844
PORTFOLIO	0.171778

Iteration 27

Method	Max Value
UCB	0.190648

EI	0.000000
PI	0.000000
THOMPSON	0.214188
KG	0.000000
ENTROPY	0.744483
PORTFOLIO	0.150193

Iteration 28

Method	Max Value
UCB	0.193486
EI	0.000000
PI	0.000000
THOMPSON	0.193254
KG	0.000000
ENTROPY	0.749605
PORTFOLIO	0.155258

Iteration 29

Method	Max Value
UCB	0.183862
EI	0.000000
PI	0.000000
THOMPSON	0.232316
KG	0.000000
ENTROPY	0.761763
PORTFOLIO	0.144440

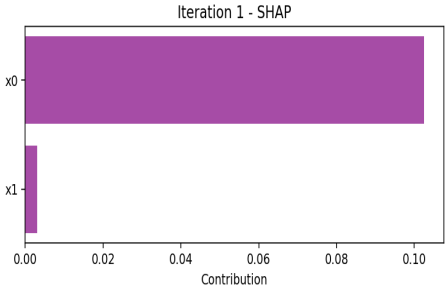
Iteration 30

Method	Max Value
UCB	0.184646
EI	0.000000
PI	0.000000
THOMPSON	0.203317
KG	0.000000
ENTROPY	0.736011

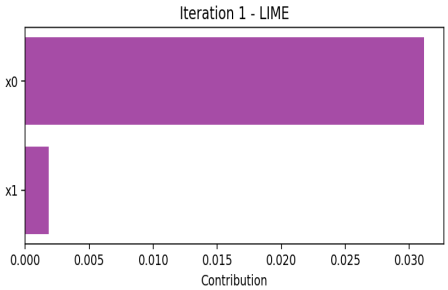
PORTFOLIO	0.128309

Feature Explanations (SHAP / LIME)

Iteration 1 - SHAP



Iteration 1 - LIME



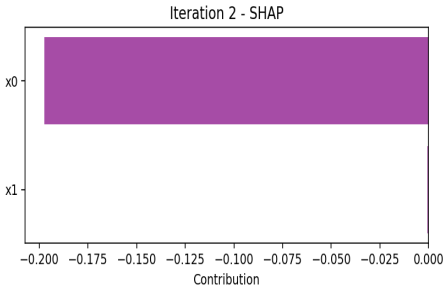
Iteration 1 - SHAP Values

Feature	SHAP Value
x0	0.10245
x1	0.00310

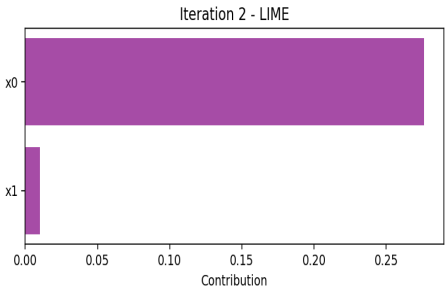
Iteration 1 - LIME Values

Feature	LIME Value
x0	0.03113
x1	0.00184

Iteration 2 - SHAP



Iteration 2 - LIME



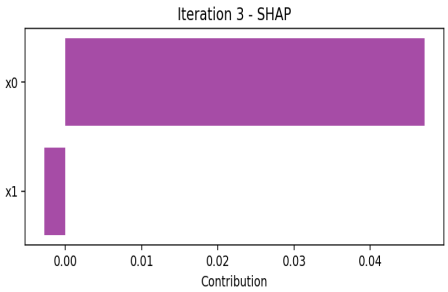
Iteration 2 - SHAP Values

Feature	SHAP Value
x0	-0.19730
x1	-0.00056

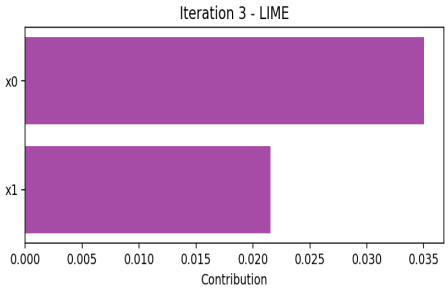
Iteration 2 - LIME Values

Feature	LIME Value
x0	0.27614
x1	0.01025

Iteration 3 - SHAP



Iteration 3 - LIME



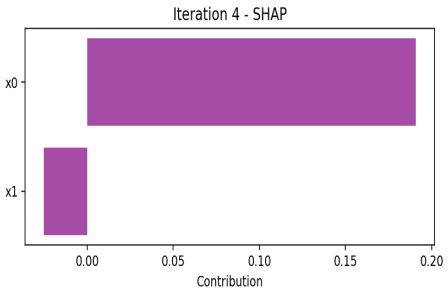
Iteration 3 - SHAP Values

Feature	SHAP Value
x0	0.04714
x1	-0.00277

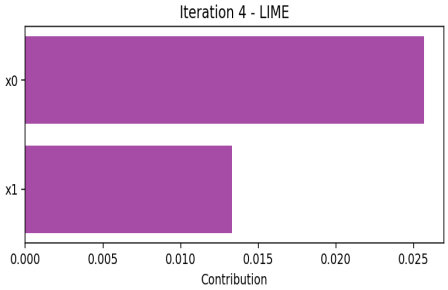
Iteration 3 - LIME Values

Feature	LIME Value
x0	0.03501
x1	0.02153

Iteration 4 - SHAP



Iteration 4 - LIME



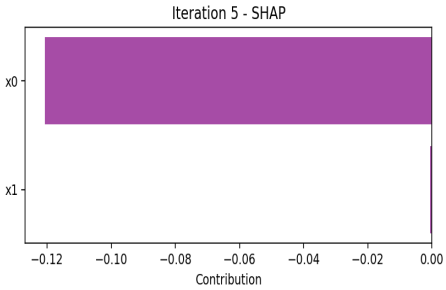
Iteration 4 - SHAP Values

Feature	SHAP Value
x0	0.19087
x1	-0.02517

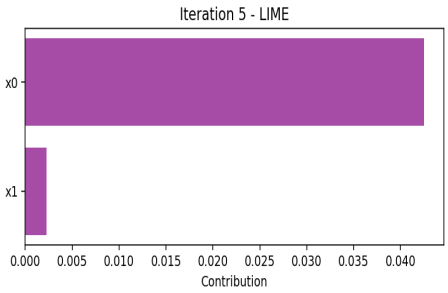
Iteration 4 - LIME Values

Feature	LIME Value
x0	0.02568
x1	0.01332

Iteration 5 - SHAP



Iteration 5 - LIME



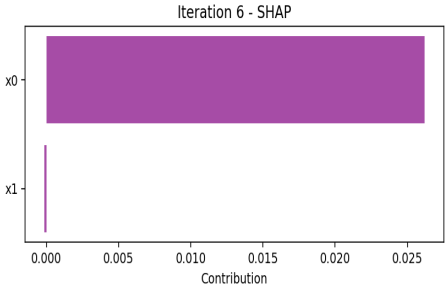
Iteration 5 - SHAP Values

Feature	SHAP Value
x0	-0.12078
x1	-0.00060

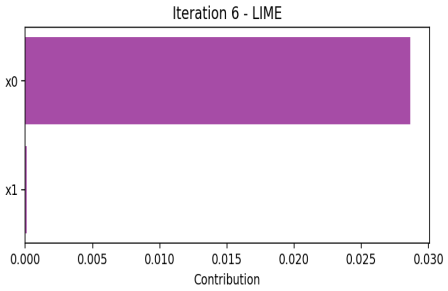
Iteration 5 - LIME Values

Feature	LIME Value
x0	0.04249
x1	0.00228

Iteration 6 - SHAP



Iteration 6 - LIME



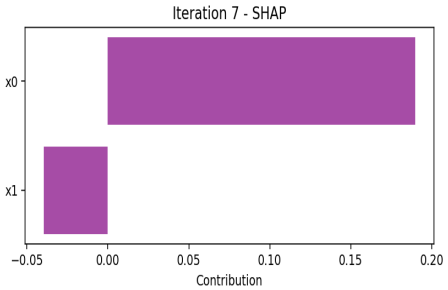
Iteration 6 - SHAP Values

Feature	SHAP Value
x0	0.02621
x1	-0.00014

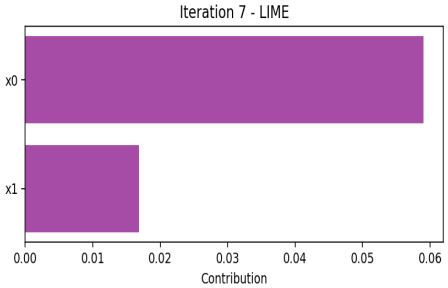
Iteration 6 - LIME Values

Feature	LIME Value
x0	0.02865
x1	0.00010

Iteration 7 - SHAP



Iteration 7 - LIME



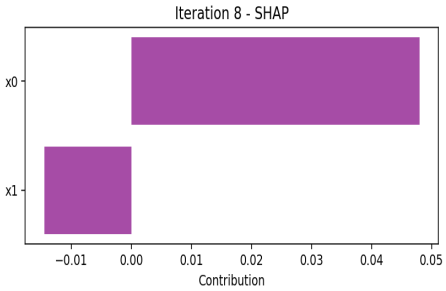
Iteration 7 - SHAP Values

Feature	SHAP Value
x0	0.18987
x1	-0.03949

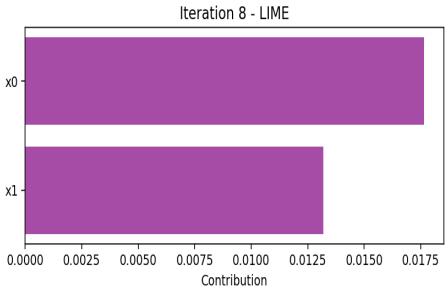
Iteration 7 - LIME Values

Feature	LIME Value
x0	0.05901
x1	0.01688

Iteration 8 - SHAP



Iteration 8 - LIME



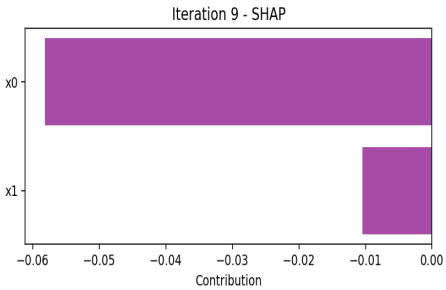
Iteration 8 - SHAP Values

Feature	SHAP Value
x0	0.04800
x1	-0.01455

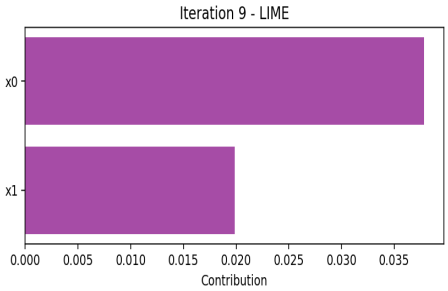
Iteration 8 - LIME Values

Feature	LIME Value
x0	0.01764
x1	0.01319

Iteration 9 - SHAP



Iteration 9 - LIME



Iteration 9 - SHAP Values

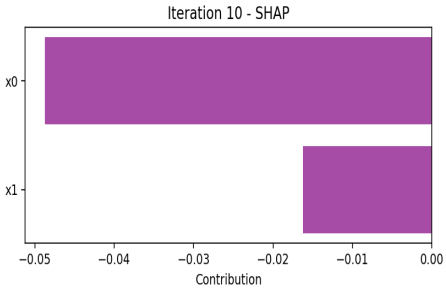
Feature	SHAP Value
x0	-0.05820
x1	-0.01045

Iteration 9 - LIME Values

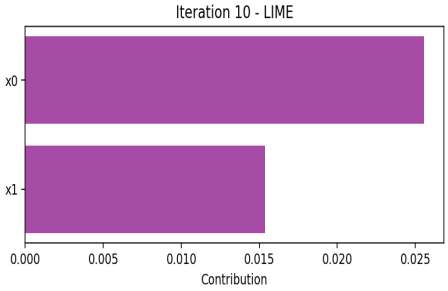
Feature	LIME Value
x0	0.03780

x1	0.01984

Iteration 10 - SHAP



Iteration 10 - LIME



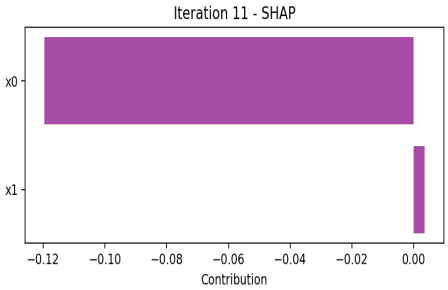
Iteration 10 - SHAP Values

Feature	SHAP Value
x0	-0.04876
x1	-0.01621

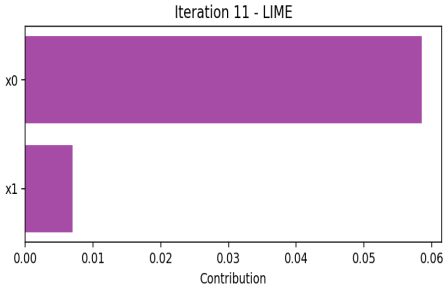
Iteration 10 - LIME Values

Feature	LIME Value
x0	0.02555
x1	0.01537

Iteration 11 - SHAP



Iteration 11 - LIME



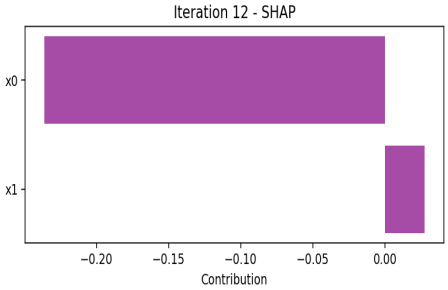
Iteration 11 - SHAP Values

Feature	SHAP Value
x0	-0.11964
x1	0.00357

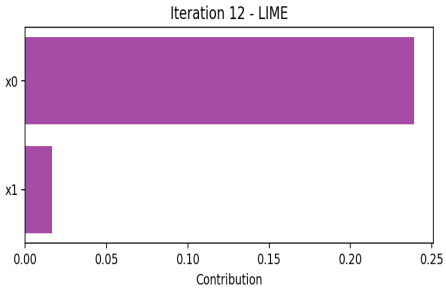
Iteration 11 - LIME Values

Feature	LIME Value
x0	0.05853
x1	0.00697

Iteration 12 - SHAP



Iteration 12 - LIME



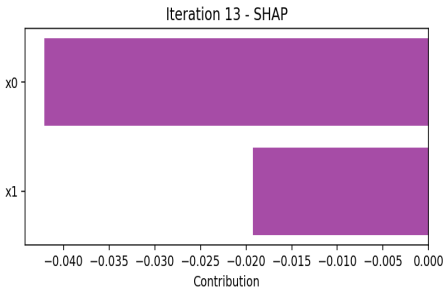
Iteration 12 - SHAP Values

Feature	SHAP Value
x0	-0.23613
x1	0.02768

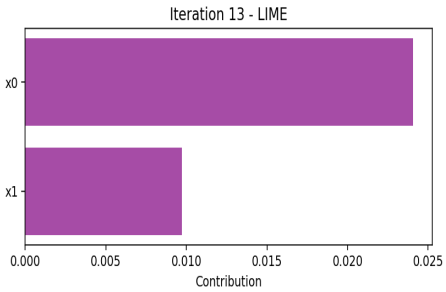
Iteration 12 - LIME Values

Feature	LIME Value
x0	0.23902
x1	0.01632

Iteration 13 - SHAP



Iteration 13 - LIME



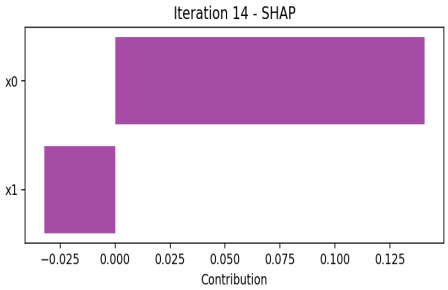
Iteration 13 - SHAP Values

Feature	SHAP Value
x0	-0.04208
x1	-0.01928

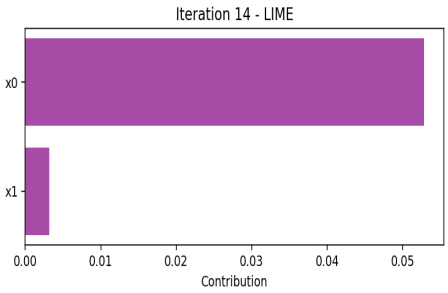
Iteration 13 - LIME Values

Feature	LIME Value
x0	0.02406
x1	0.00971

Iteration 14 - SHAP



Iteration 14 - LIME



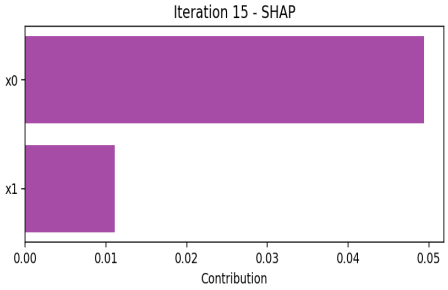
Iteration 14 - SHAP Values

Feature	SHAP Value
x0	0.14082
x1	-0.03219

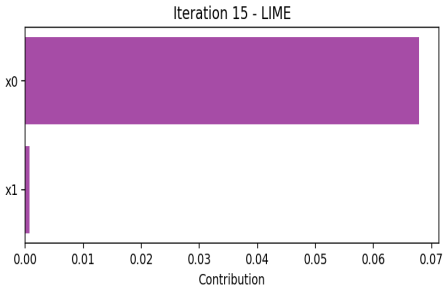
Iteration 14 - LIME Values

Feature	LIME Value
x0	0.05283
x1	0.00318

Iteration 15 - SHAP



Iteration 15 - LIME



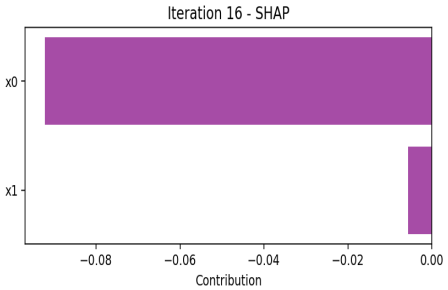
Iteration 15 - SHAP Values

Feature	SHAP Value
x0	0.04938
x1	0.01105

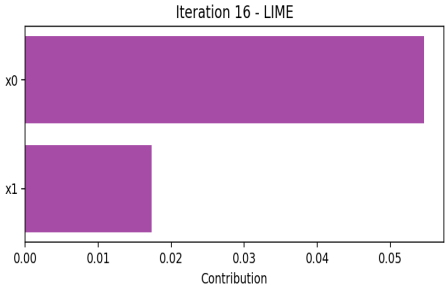
Iteration 15 - LIME Values

Feature	LIME Value
x0	0.06788
x1	0.00072

Iteration 16 - SHAP



Iteration 16 - LIME



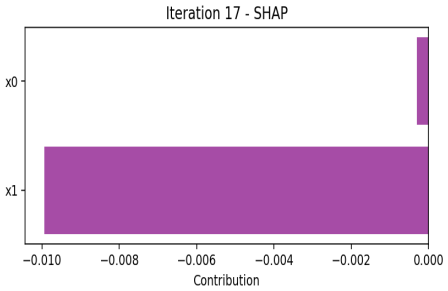
Iteration 16 - SHAP Values

Feature	SHAP Value
x0	-0.09222
x1	-0.00563

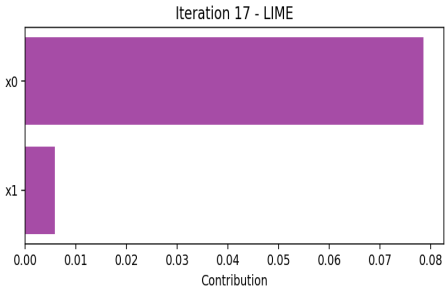
Iteration 16 - LIME Values

Feature	LIME Value
x0	0.05460
x1	0.01728

Iteration 17 - SHAP



Iteration 17 - LIME



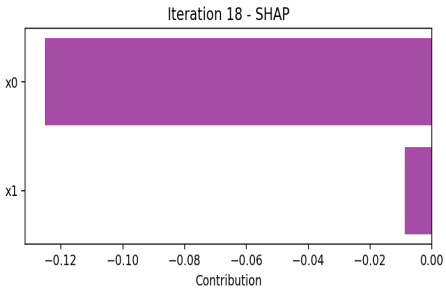
Iteration 17 - SHAP Values

Feature	SHAP Value
x0	-0.00031
x1	-0.00994

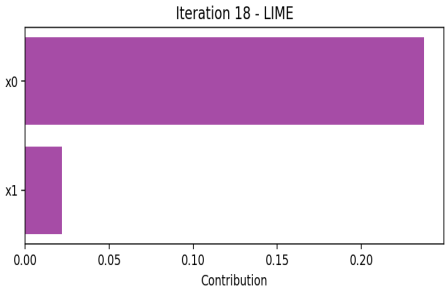
Iteration 17 - LIME Values

Feature	LIME Value
x0	0.07863
x1	0.00581

Iteration 18 - SHAP



Iteration 18 - LIME



Iteration 18 - SHAP Values

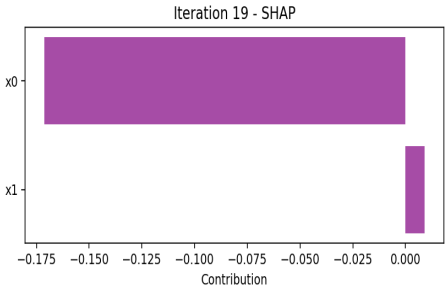
Feature	SHAP Value
x0	-0.12525
x1	-0.00881

Iteration 18 - LIME Values

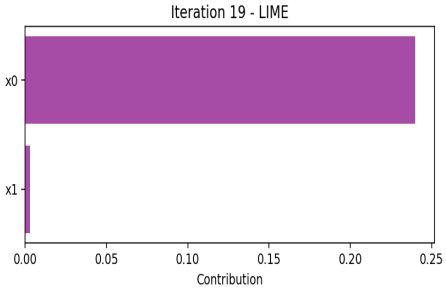
Feature	LIME Value
x0	0.23731

x1	0.02170

Iteration 19 - SHAP



Iteration 19 - LIME



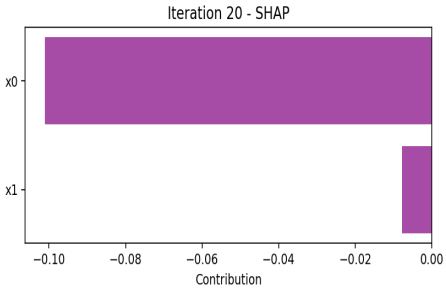
Iteration 19 - SHAP Values

Feature	SHAP Value
x0	-0.17125
x1	0.00911

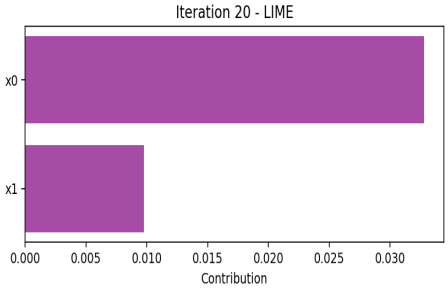
Iteration 19 - LIME Values

Feature	LIME Value
x0	0.23992
x1	0.00274

Iteration 20 - SHAP



Iteration 20 - LIME



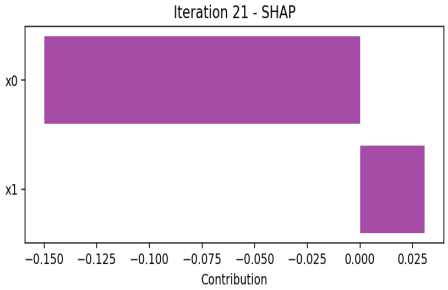
Iteration 20 - SHAP Values

Feature	SHAP Value
x0	-0.10111
x1	-0.00789

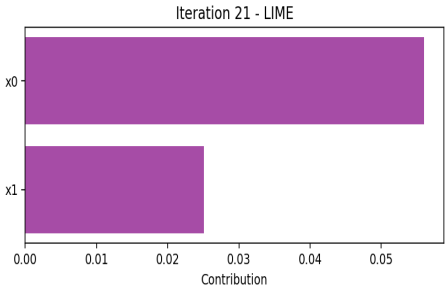
Iteration 20 - LIME Values

Feature	LIME Value
x0	0.03279
x1	0.00977

Iteration 21 - SHAP



Iteration 21 - LIME



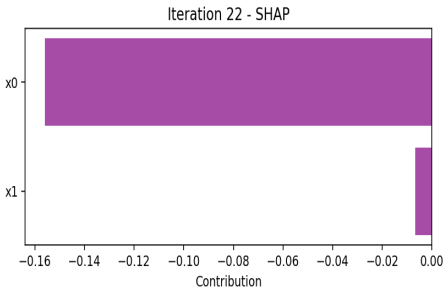
Iteration 21 - SHAP Values

Feature	SHAP Value
x0	-0.14994
x1	0.03074

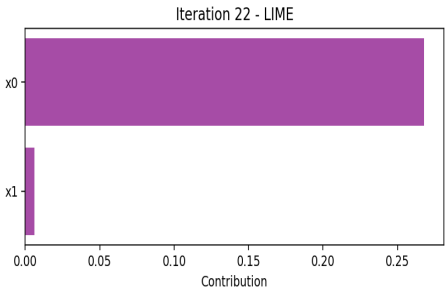
Iteration 21 - LIME Values

Feature	LIME Value
x0	0.05598
x1	0.02512

Iteration 22 - SHAP



Iteration 22 - LIME



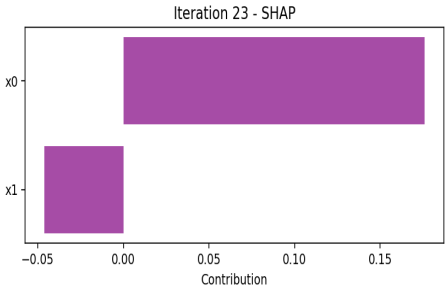
Iteration 22 - SHAP Values

Feature	SHAP Value
x0	-0.15603
x1	-0.00673

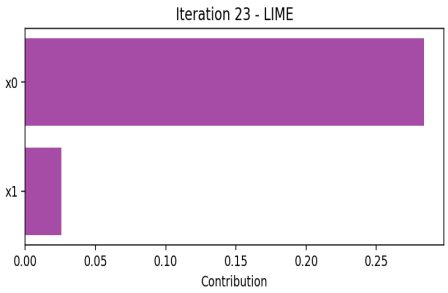
Iteration 22 - LIME Values

Feature	LIME Value
x0	0.26780
x1	0.00607

Iteration 23 - SHAP



Iteration 23 - LIME



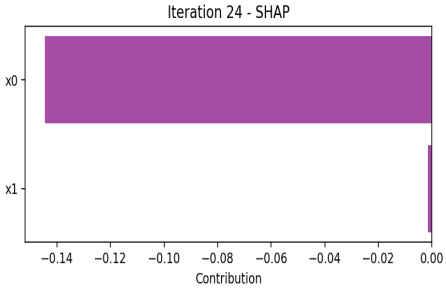
Iteration 23 - SHAP Values

Feature	SHAP Value
x0	0.17608
x1	-0.04618

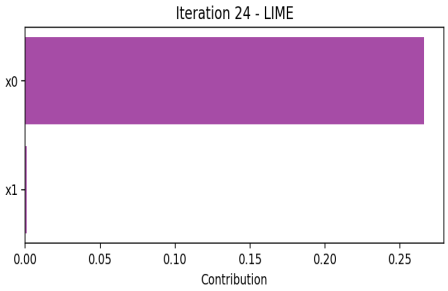
Iteration 23 - LIME Values

Feature	LIME Value
x0	0.28379
x1	0.02577

Iteration 24 - SHAP



Iteration 24 - LIME



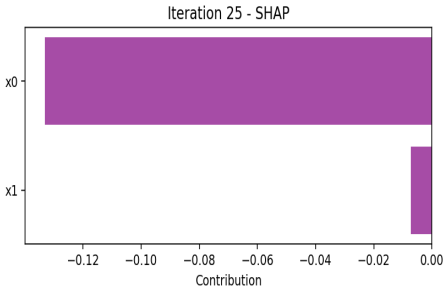
Iteration 24 - SHAP Values

Feature	SHAP Value
x0	-0.14455
x1	-0.00161

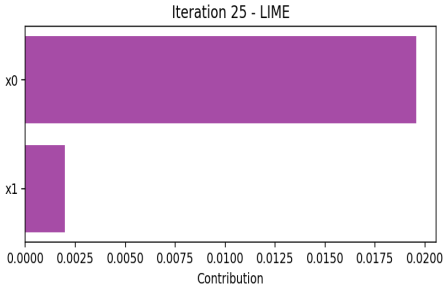
Iteration 24 - LIME Values

Feature	LIME Value
x0	0.26596
x1	0.00085

Iteration 25 - SHAP



Iteration 25 - LIME



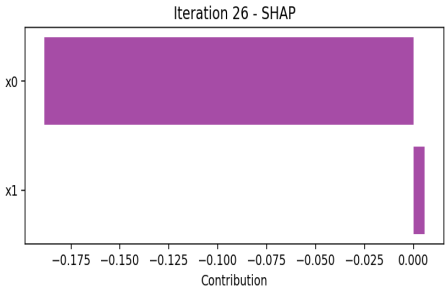
Iteration 25 - SHAP Values

Feature	SHAP Value
x0	-0.13320
x1	-0.00719

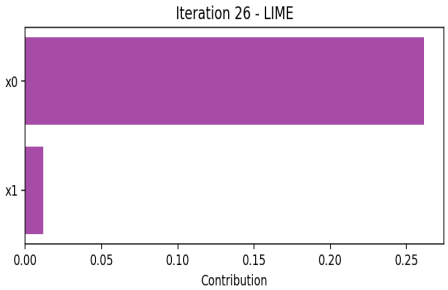
Iteration 25 - LIME Values

Feature	LIME Value
x0	0.01957
x1	0.00198

Iteration 26 - SHAP



Iteration 26 - LIME



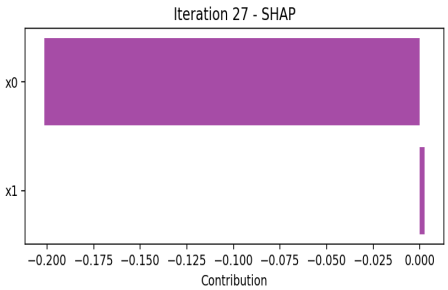
Iteration 26 - SHAP Values

Feature	SHAP Value
x0	-0.18866
x1	0.00586

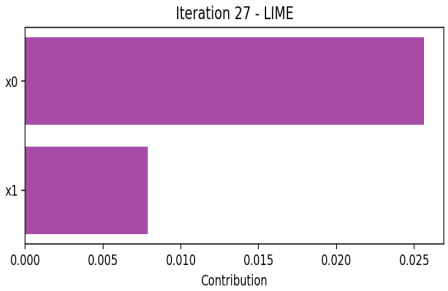
Iteration 26 - LIME Values

Feature	LIME Value
x0	0.26133
x1	0.01156

Iteration 27 - SHAP



Iteration 27 - LIME



Iteration 27 - SHAP Values

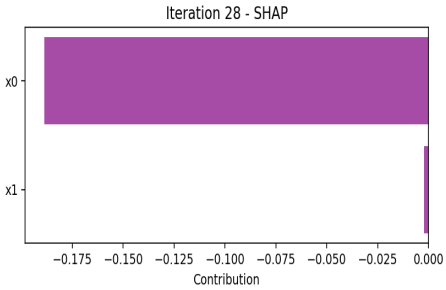
Feature	SHAP Value
x0	-0.20142
x1	0.00265

Iteration 27 - LIME Values

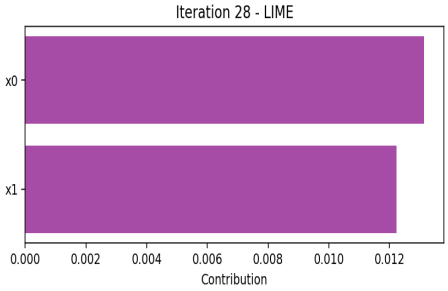
Feature	LIME Value
x0	0.02562

x1	0.00787

Iteration 28 - SHAP



Iteration 28 - LIME



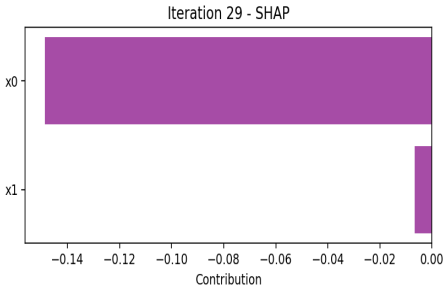
Iteration 28 - SHAP Values

Feature	SHAP Value
x0	-0.18869
x1	-0.00234

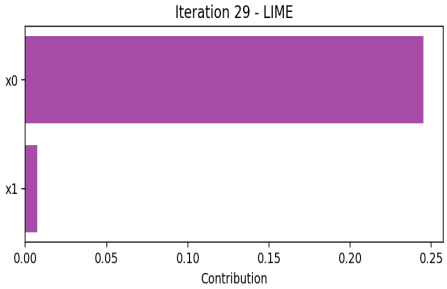
Iteration 28 - LIME Values

Feature	LIME Value
x0	0.01313
x1	0.01224

Iteration 29 - SHAP



Iteration 29 - LIME



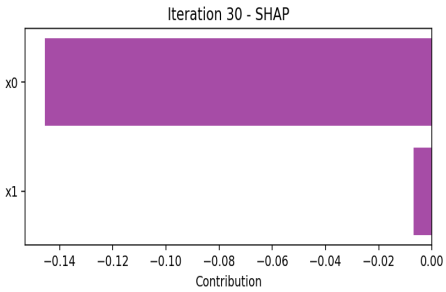
Iteration 29 - SHAP Values

Feature	SHAP Value
x0	-0.14879
x1	-0.00675

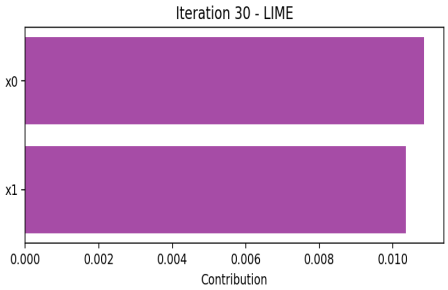
Iteration 29 - LIME Values

Feature	LIME Value
x0	0.24504
x1	0.00747

Iteration 30 - SHAP



Iteration 30 - LIME



Iteration 30 - SHAP Values

Feature	SHAP Value
x0	-0.14560
x1	-0.00701

Iteration 30 - LIME Values

Feature	LIME Value
x0	0.01085
x1	0.01036